

内容提要

- Ensemble数据库简介
- Ensemble数据查询
- 相关工具使用



Ensemble数据库简介

- Ensemble是由英国的Sanger研究所以及欧洲生物信息学研究所共同协作开发的数据库
- 最主要的功能是在于对真核生物基因组的自动注释:
 - Regulatory regions
 - Conserved base pairs across species
 - Sequence variations
- 该数据库的大部分软件都是由Perl编写的，在BioPerl的基础框架下开发，并提供Perl的API接口

数据来源

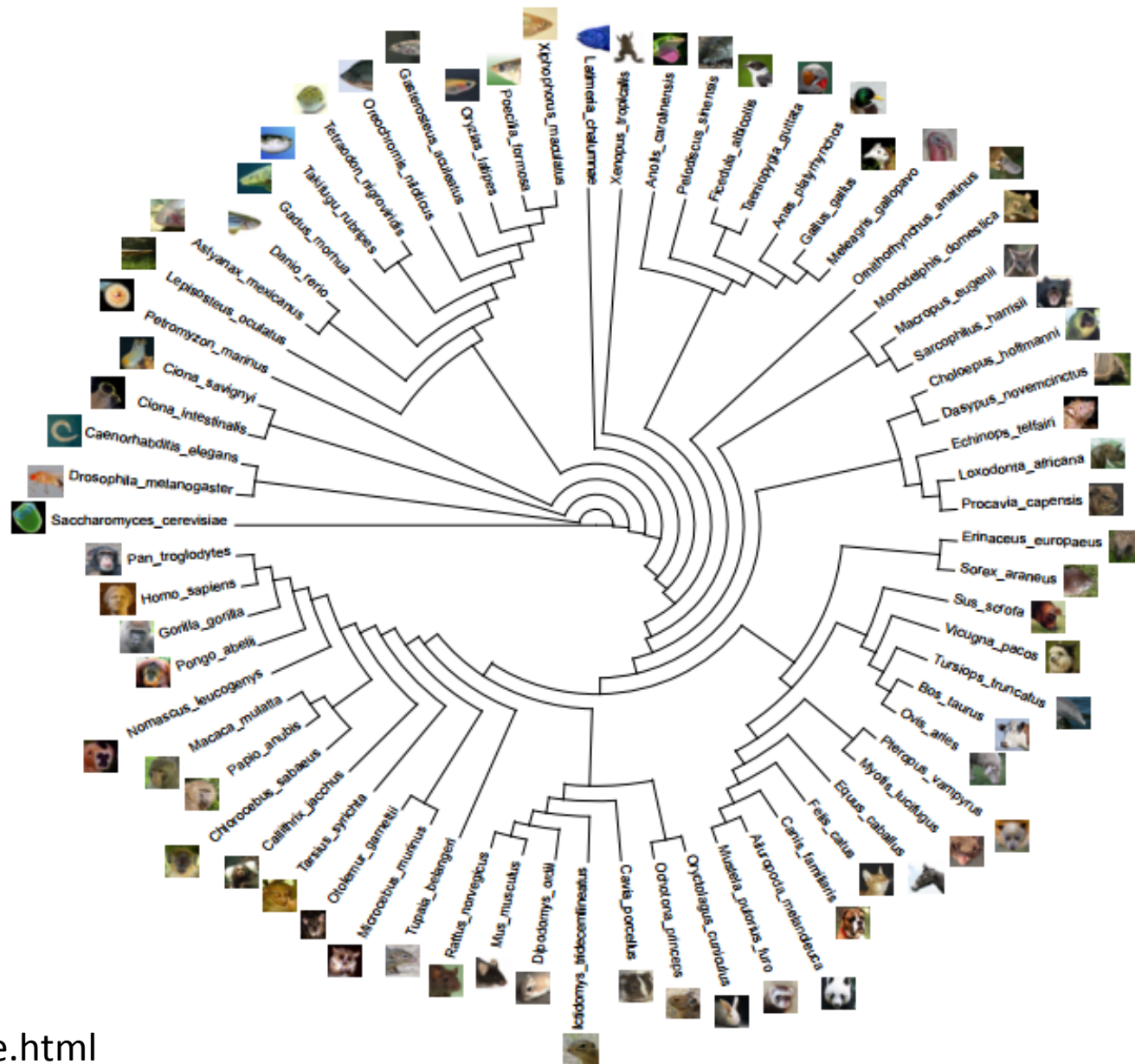
- based on protein and mRNA evidence in UniProtKB and NCBI RefSeq databases, along with manual annotation from the VEGA/Havana group

网站结构

- 主站点：脊椎动物
- 其余站点：
 - 非脊椎动物（insect vectors, nematodes）：
<http://metazoa.ensembl.org/index.html>
 - 植物 和 真菌：
 - <http://plants.ensembl.org/index.html>
 - <http://fungi.ensembl.org/index.html>
 - 原生动物，细菌：
 - <http://protists.ensembl.org/>
 - <http://bacteria.ensembl.org/>

网站结构

- 主站点：脊椎动物
- 目前为止一共69个物种
- 囊括了：
 - 鱼类
 - 爬行类
 - 鸟类
 - 哺乳类
 - 两栖类
 - ...



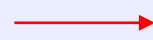
<http://asia.ensembl.org/info/about/speciestree.html>

Ensembl主页

<http://www.ensembl.org/index.html>



关键词搜索框



Search: for
e.g. BRCA2 or rat 5:62797383-63627669 or rs699 or coronary heart disease

常用
物种
快速
导航

Browse a Genome

The Ensembl project produces genome databases for vertebrates and other eukaryotic species, and makes this information freely available online.

Popular genomes

**Human**
GRCh38.p5

**Human**
GRCh37

**Mouse**
GRCm38.p4

**Zebrafish**
GRCz10

★ [Log in to customize this list](#)

All genomes

-- Select a species --

[View full list of all Ensembl species](#)

Other species are available in [Ensembl Pre!](#) and [Ensembl Genomes](#)

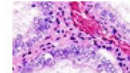
Still using Human GRCh37?

Go to 

Variant Effect Predictor



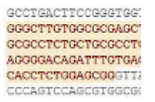
Gene expression in different tissues



Find SNPs and other variants for my gene



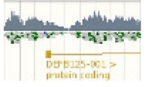
Retrieve gene sequence



Compare genes across species



Use my own data in Ensembl



ENCODE data in Ensembl



What's New in Ensembl Release 84 (March 2016)

- 20 haematopoietic primary cell epigenomes from the BLUEPRINT project
- Mouse: update to Ensembl-Havana GENCODE gene set
- Track hub registry interface
- dbSNP 146 for Human, Cow and Dog
- Pairwise LD calculation on LD variant page

[Full details](#) | [All web updates, by release](#) | [More news on our blog](#)

- 02 Jun 2016: [What's coming in Ensembl release 85](#)
- 25 Apr 2016: [DNA day and Malaria day: a story of scientific endeavour](#)
- 31 Mar 2016: [Ensembl 85 and Ensembl Genomes 32](#)

[Go to Ensembl blog](#)

[Recent tweets from @ensembl](#)

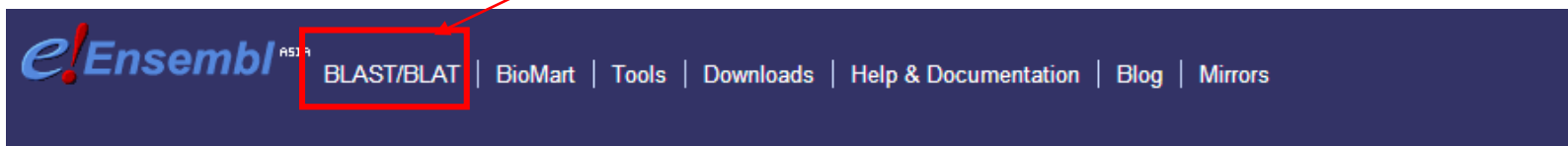
新闻通知栏

常用功能



Ensemble检索

方法三:



方法一:

Search: for

e.g. BRCA2 or rat 5:62797383-63627669 or coronary heart disease

物种

检索词

方法二:

Browse a Genome

The Ensembl project produces genome databases for vertebrates and other eukaryotic species, and makes this information freely available online.

Popular genomes



Human
GRCh38.p2



Mouse
GRCm38.p3



Zebrafish
GRCz10

★ [Log in to customize this list](#)

All genomes

-- Select a species --

[View full list of all Ensembl species](#)

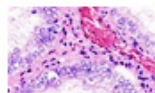
ENCODE data in Ensembl



Variant Effect Predictor



Gene expression in different tissues



Find SNPs and other variants for my gene

GTATACATTC
CTTAAAGTCTT
CTTCTAATTCT
GATCTTTTTC

Retrieve gene sequence

GCCTGACTTCGGGTGG/
GGGCTTGTGGGGGAGC/
GCGCCTCTGCTGCGCCT/
AGGGACAGATTGTGA/
CACCTCTGGAGCGGTT/
CCCACTCCAGCGTGGCG/

Compare genes across species

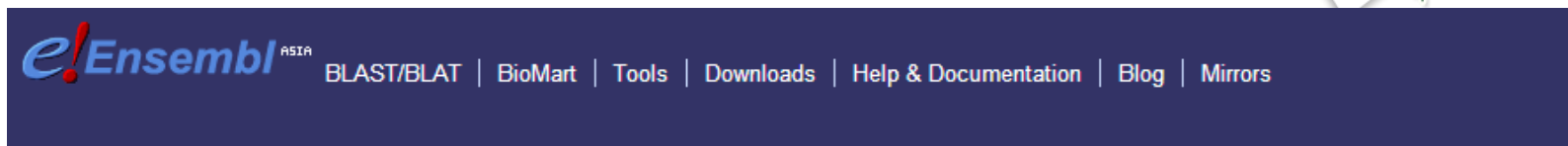


[Use my own data in](#)

[Learn about a disease or](#)

检索方法一：检索词

- 1. 基因名称
- 2. 染色体位置
- 3. 疾病名称



方法一：→ Search: for

e.g. BRCA2 or rat 5:62797383-63627669 or coronary heart disease

Browse a Genome

The Ensembl project produces genome databases for vertebrates and other eukaryotic species, and makes this information freely available online.

Popular genomes



Human
GRCh38.p2



Mouse
GRCm38.p3



Zebrafish
GRCz10

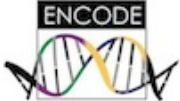
★ [Log in to customize this list](#)

All genomes

-- Select a species --

[View full list of all Ensembl species](#)

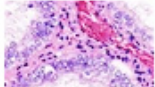
ENCODE data in Ensembl



Variant Effect Predictor



Gene expression in different tissues



Find SNPs and other variants for my gene

```

GTTTATACATTCT
CTTAAAGTCTT
CTTCTAATTCT
CTTCTAATTCT

```

Retrieve gene sequence

```

GCTGACTTCGGGTGG
GGGCTTGTGGCGGAGC
GCGCTCTGCTGGCCT
AGGGGACAGATTGTGA
CACCTCTGGAGCGGTT
CCAGTCCAGCGTGGCG

```

Compare genes across species



Use my own data in

Learn about a disease or

检索方法二：直接点击物种名链接



[BLAST/BLAT](#) | [BioMart](#) | [Tools](#) | [Downloads](#) | [Help & Documentation](#) | [Blog](#) | [Mirrors](#)

Search all species...

[Login/Register](#)

Human (GRCh38.p5) ▾



Human

Homo sapiens

Search all categories ▾ Search Human...

e.g. [BRCA2](#) or [17:63973115-64437414](#) or [rs1333049](#) or [osteoarthritis](#)

Genome assembly: GRCh38.p5 (GCA_000001405.20)

 More information and statistics

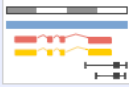
 Download DNA sequence (FASTA)

 Convert your data to GRCh38 coordinates

 Display your data in Ensembl

Other assemblies

GRCh37 Full Feb 2014 archive with BLAST, VEP and BioMart ▾

[View karyotype](#)[Example region](#)

Gene annotation

What can I find? Protein-coding and non-coding genes, splice variants, cDNA and protein sequences, non-coding RNAs.

 More about this genebuild, including RNASeq gene expression models

 Download genes, cDNAs, ncRNA, proteins (FASTA)

 Update your old Ensembl IDs

 Additional manual annotation can be found in Vega

[Example gene](#)[Example transcript](#)

Comparative genomics

What can I find? Homologues, gene trees, and whole genome alignments across multiple species.

 More about comparative analysis

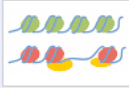
 Download alignments (EMF)

[Example gene tree](#)

Regulation

What can I find? DNA methylation, transcription factor binding sites, histone modifications, and regulatory features such as enhancers and repressors, and microarray annotations.

 More about the Ensembl regulatory build and microarray annotation

[Example regulatory](#)

Variation

What can I find? Short sequence variants and longer structural variants; disease and other phenotypes

 More about variation in Ensembl

 Download all variants (GVF)

 Variant Effect Predictor 

[Example variant](#)[Example phenotype](#)

检索方法三：序列相似性查找 (BLAST/BLAT)



<http://asia.ensembl.org/Multi/Tools/Blast?db=core>

The screenshot shows the Ensembl BLAST/BLAT search page. The top navigation bar includes links for BLAST/BLAT, BioMart, Tools, Downloads, Help & Documentation, Blog, and Mirrors. A search bar for species is also present. The left sidebar contains a 'Web Tools' menu with options like BLAST/BLAT, Variant Effect Predictor, and Assembly Converter, along with buttons for 'Configure this page', 'Add your data', 'Export data', 'Share this page', and 'Bookmark this page'. The main content area is titled 'BLAST/BLAT search' and includes a 'Create new ticket' section. Below this, there is a message about using the GRCh37 website. The 'Sequence data' section features a text input field for sequences (up to 30), an option to upload a file, and radio buttons for DNA or Protein. The 'Search against' section shows a dropdown menu set to 'Human (Homo sapiens)' and a 'DNA database' dropdown set to 'Genomic sequence'.

Ensembl BLAST/BLAT | BioMart | Tools | Downloads | Help & Documentation | Blog | Mirrors

Search all species...

Species BLAST/BLAT

Web Tools

- Web Tools
- BLAST/BLAT
- Variant Effect Predictor
- Assembly Converter

Configure this page

Add your data

Export data

Share this page

Bookmark this page

BLAST/BLAT search

Create new ticket:

BLAST/BLAT for Human GRCh37

If you are looking for BLAST/BLAT for Human GRCh37, please go to [GRCh37 website](#).

Sequence data:

Maximum of 30 sequences (type in plain text, FASTA or sequence ID)

Or upload sequence file 选择文件 未选择文件

DNA

Protein

Search against:

Human (Homo sapiens)

Type in to add a species...

DNA database Genomic sequence

检索-基因名

单击这里



结果分类

Ensembl ASIA BLAST/BLAT | BioMart | Tools | Downloads | Help & Documentation | Blog | Mirrors

Species ▾

Search all species...

Restrict category to:

Gene	365
Transcript	616
Phenotype	3
Somatic Mutation	1275
GeneTree	60
Clones & Regions	1
Marker	6
ProbeFeature	243
Help & Docs	11
Protein Domain	172
... 2 more categories ...	

Restrict species to:

Human	12185
Mouse	1535
Zebrafish	28
Alpaca	21
Amazon molly	26
Anole lizard	22
... 63 more species ...	

Per page: 10 25 50 100

Layout: Standard Table

BRCA2

20070 results match BRCA2

Best gene match

Human Gene

BRCA2

Protein coding gene

breast cancer 2

HGNC Symbol; Acc:HGNC:1101

Suggestions

arca2	bdca2	brca1	brca1	prca2	brovca2	dmbrc2	a3rca2
a6rca2	b0eca2	b0wca2	b0yca2	b1wca2	b3fca2	b3gca2	
b3mca2	b4gca2	b4ica2	b4pca2	b5aca2	b6aca2	b7tca2	b7uca2
b8mca2	brcaa1	d6rca2	e6rca2	f6rca2	g1rca2	g3rca2	h2rca2
n1rca2	s4rca2	s9rca2	serca2	abca2	bra	brac	braf
brag2	brak	bral2	bran	brap	brat	broox	brna
							odca2

BRCA2 (Human Gene)

ENSG00000139618 13:32315474-32400266:1

Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101]

BRCA2 (Vega gene) is associated with Gene ENSG00000139618

Variant table • Phenotypes • Location • External Refs. • Regulation • Orthologues • Gene tree

BRCA2-001 (Human Transcript)

ENST00000380152 13:32315474-32400266:1

Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101]

BRCA2-001 (Vega transcript) is associated with Transcript ENST00000380152

Location • External Refs. • cDNA seq. • Exons • Variant table • Protein seq. • Population • Protein summary

BRCA2-201 (Human Transcript)

ENST00000544455 13:32315480-32399668:1

Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101]

LRG_293t1 (LRG display in Ensembl record with a description of Locus Reference Genomic record for BRCA2) is associated with Transcript ENST00000544455

Location • External Refs. • cDNA seq. • Exons • Variant table • Protein seq. • Population • Protein summary

BRCA2-004 (Human Transcript)

ENST00000614259 13:32316461-32362659:1

Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101]

BRCA2-004 (Vega transcript) is associated with Transcript ENST00000614259

Location • External Refs. • cDNA seq. • Exons • Variant table • Population

BRCA2-005 (Human Transcript)

ENST00000528762 13:32370971-32379495:1

Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101]

BRCA2-005 (Vega transcript) is associated with Transcript ENST00000528762

Location • External Refs. • cDNA seq. • Exons • Variant table • Protein seq. • Population • Protein summary

BRCA2-003 (Human Transcript)

ENST00000530893 13:32315505-32333291:1

Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101]

所有相关的搜索结果列表

调整每页
显示的结果数

Layout: Standard Table

同样的检索关键词，不同的结果显示方式。

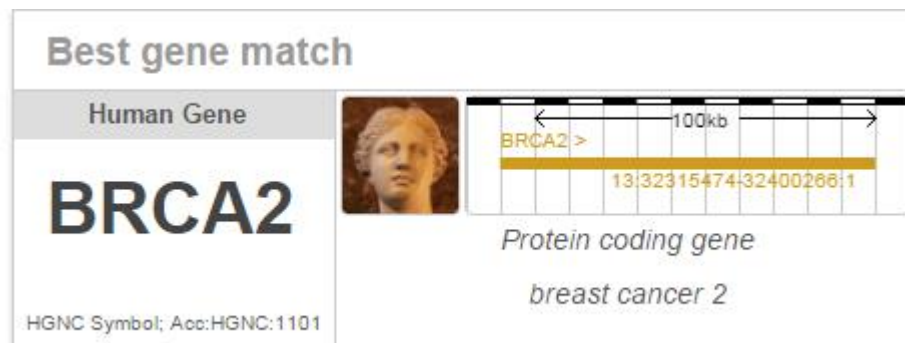


Show 10 entries
Show/hide columns
BRCA2

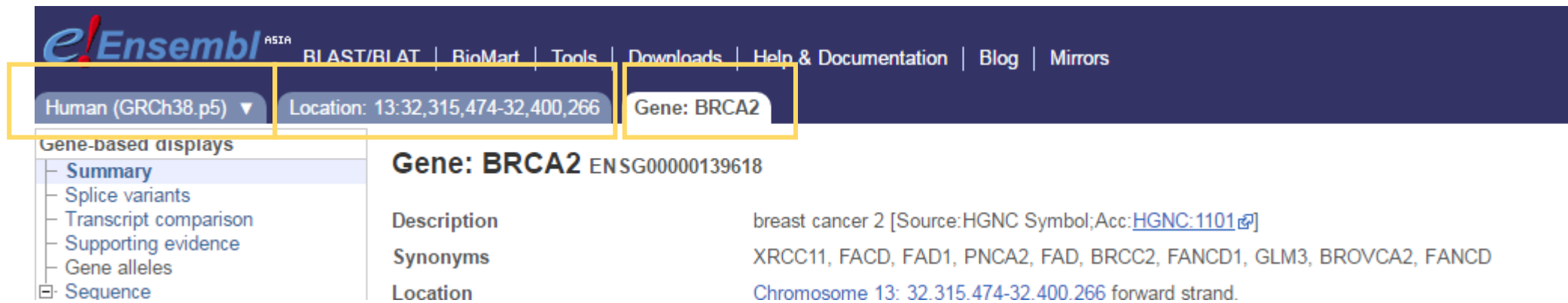
ID	Name	Species	Category	Description
ENSG00000139618	BRCA2	Human	Gene	Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101] <i>BRCA2</i> (Vega gene) is associated with Gene ENSG00000139618
ENST00000380152	BRCA2-001	Human	Transcript	Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101] <i>BRCA2-001</i> (Vega transcript) is associated with Transcript ENST00000380152
ENST00000544455	BRCA2-201	Human	Transcript	Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101] LRG_293t1 (LRG display in Ensembl record with a description of Locus Reference Genomic record for <i>BRCA2</i>) is associated with Transcript ENST00000544455
ENST00000614259	BRCA2-004	Human	Transcript	Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101] <i>BRCA2-004</i> (Vega transcript) is associated with Transcript ENST00000614259
ENST00000528762	BRCA2-005	Human	Transcript	Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101] <i>BRCA2-005</i> (Vega transcript) is associated with Transcript ENST00000528762
ENST00000530893	BRCA2-003	Human	Transcript	Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101] <i>BRCA2-003</i> (Vega transcript) is associated with Transcript ENST00000530893
ENST00000533776	BRCA2-006	Human	Transcript	Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101] <i>BRCA2-006</i> (Vega transcript) is associated with Transcript ENST00000533776
ENST00000470094	BRCA2-002	Human	Transcript	Breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101] <i>BRCA2-002</i> (Vega transcript) is associated with Transcript ENST00000470094
ENSMUSG00000041147	Brca2	Mouse	Gene	Breast cancer 2, early onset [Source:MGI Symbol;Acc:MGI:109337] <i>Brca2</i> (Vega gene) is associated with Gene ENSMUSG00000041147
ENSMUST00000044620	Brca2-201	Mouse	Transcript	Breast cancer 2, early onset [Source:MGI Symbol;Acc:MGI:109337] <i>Brca2-201</i> (MGI transcript name record with a description of breast cancer 2, early onset) is associated with Transcript ENSMUST00000044620

<< < 1 2 3 4 5 6 7 8 ... 2006 2007 > >>

具体例子



每个检索结果页都有3个tab



Ensembl^{ASIA} | BLAST/BLAT | BioMart | Tools | Downloads | Help & Documentation | Blog | Mirrors

Human (GRCh38.p5) ▼ | Location: 13:32,315,474-32,400,266 | Gene: BRCA2

Gene-based displays

- Summary
- Splice variants
- Transcript comparison
- Supporting evidence
- Gene alleles
- Sequence

Gene: BRCA2 ENSG00000139618

Description: breast cancer 2 [Source:HGNC Symbol;Acc:[HGNC:1101](#)]

Synonyms: XRCC11, FACD, FAD1, PNCA2, FAD, BRCC2, FANCD1, GLM3, BROVCA2, FANCD

Location: Chromosome 13: 32.315.474-32.400.266 forward strand.

Tab 1: 物种切换

Human (GRCh38.p5) ▲

Location: 13:32,315,474-32,400,266

Gene: BRCA2

Favourite species

Human

Mouse

Zebrafish

All species

Alpaca

Fugu

Pika

Amazon molly

Gibbon

Platyfish

Anole lizard

Gorilla

Platypus

Armadillo

Guinea Pig

Rabbit

Bushbaby

Hedgehog

Rat

C.intestinalis

Horse

Saccharomyces cerevisiae

C.savignyi

Human

Sheep

Caenorhabditis elegans

Hyrax

Shrew

Cat

Kangaroo rat

Sloth

Cave fish

Lamprey

Spotted gar

Chicken

Lesser hedgehog tenrec

Squirrel

Chimpanzee

Macaque

Stickleback

Chinese softshell turtle

Marmoset

Tarsier

Cod

Medaka

Tasmanian devil

Coelacanth

Megabat

Tetraodon

Cow

Microbat

Tilapia

Dog

Mouse

Tree Shrew

Dolphin

Mouse Lemur

Turkey

Duck

Olive baboon

Vervet-AGM

Elephant

Opossum

Wallaby

Ferret

Orangutan

Xenopus

Flycatcher

Panda

Zebra Finch

Fruitfly

Pig

Zebrafish

螺旋

Tab 2: Genome location card

Human (GRCh38.p5)

Location: 13:32,330,879-32,339,085

Gene: BRCA2

Location-based displays

- Whole genome
- Chromosome summary
- Region overview
- Region in detail
- Comparative Genomics
 - Alignments (image)
 - Alignments (text)
 - Region Comparison
 - Synteny
- Genetic Variation
 - Resequencing
 - Linkage Data
- Markers
- Other genome browsers
 - UCSC
 - NCBI
 - Vega
 - Ensembl GRCh37

Configure this page

Add your data

Export data

Share this page

Bookmark this page

Chromosome 13: 32,330,879-32,339,085

Assembly exceptions

Chr. 13

Assembly exceptions

Region in detail

Chromosome bands

Contigs

Genes (Comprehensive set from GENCODE 24)

Gene Legend

- merged Ensembl/Havana
- pseudogene
- processed transcript
- RNA gene

31.90 Mb

32.00 Mb

32.10 Mb

32.20 Mb

32.30 Mb

32.40 Mb

32.50 Mb

32.60 Mb

32.70 Mb

32.80 Mb

q13.1

AL137143.8 >

AL138892.26 >

AL445212.9 >

AL137247.14 >

Z84467.1 >

Z75889.1 >

AL13

EEF1DP3 >

LINC01073 >

Metazoa_SRP >

FRY >

< RY1-257C22.2

< FRY-AS1

< ZAR11

BRCA2 >

IFIT1P1 >

< N4BP2L2

< N4BP2L1

< SNORA18

< RP11-298P3.4

ATP8A2P2 >

PDS5B >

< RNY1P4

RP11-31

Location: 13:32330879-32339085

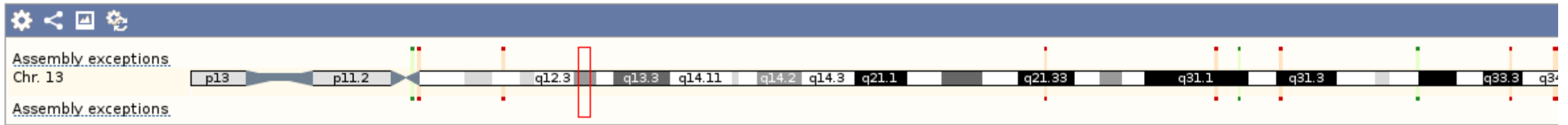
Go

Gene:

Go

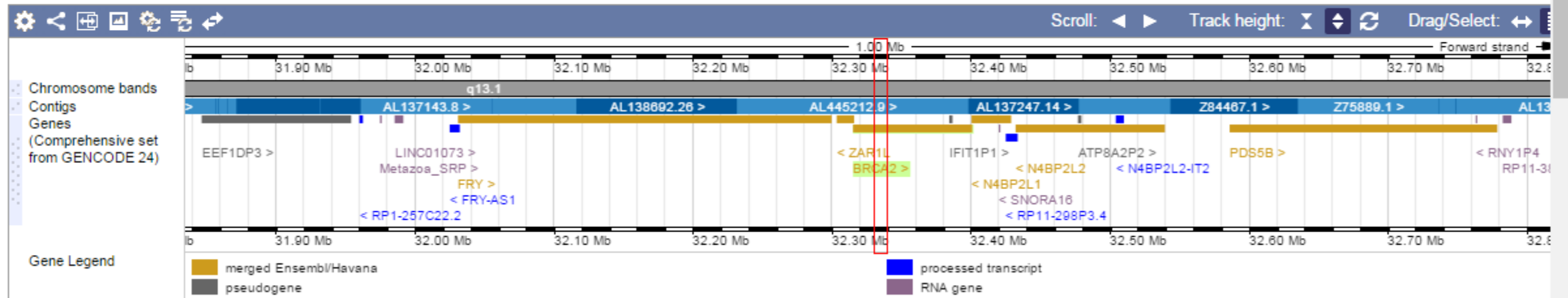
- Chromosome Summary

Chromosome 13: 32,330,879-32,339,085



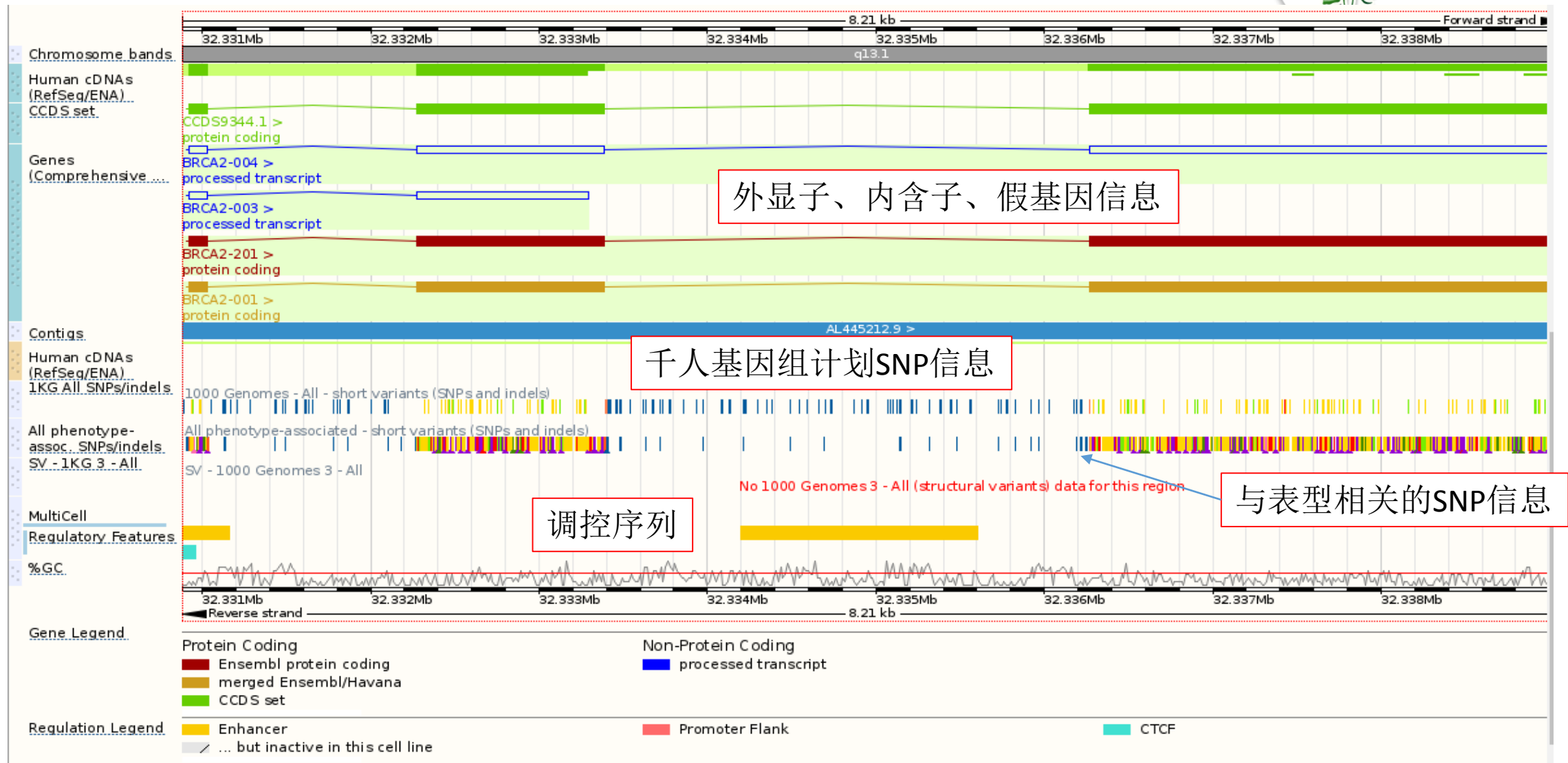
可以通过移动红色的框，移动关注的区间

Region in detail ?



- Merged Ensembl/Havana gene record
- Processed transcript
- Pseudogene
- RNA gene

Region in detail



Tab 3: Gene card

1) 基因信息的文字介绍



2) 各类相关的注释信息

Human (GRCh38.p5) Location: 13:32,315,474-32,400,266 Gene: BRCA2

Gene-based displays

Summary

Splice variants

Transcript comparison

Supporting evidence

Gene alleles

Sequence

Secondary Structure

External references

Regulation

Ontologies

GO: Biological process

GO: Molecular function

GO: Cellular component

Comparative Genomics

Genomic alignments

Gene tree

Gene gain/loss tree

Orthologues

Paralogues

Ensembl protein families

Phenotype

Genetic Variation

Variant table

Variant image

Structural variants

External data

Gene expression

ID History

Gene history

Configure this page

Add your data

Export data

Share this page

Bookmark this page

Gene: BRCA2 ENSG00000139618

Description

Synonyms

Location

About this gene

Transcripts

Summary

Name

CCDS

UniProtKB

RefSeq

LRG

Ensembl version

Other assemblies

Gene type

Annotation method

Alternative genes

Annotation Attributes

breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101]

XRCC11, FACD, FAD1, PNCA2, FAD, BRCC2, FANCD1, GLM3, BROVCA2, FANCD

Chromosome 13: 32,315,474-32,400,266 forward strand.
GRCh38:CM000675.2

This gene has 7 transcripts (splice variants), 63 orthologues, is a member of 1 Ensembl protein family and is associated with 51 phenotypes.

Show transcript table

BRCA2 (HGNC Symbol)

This gene is a member of the Human CCDS set: CCDS9344.1

This gene has proteins that correspond to the following Uniprot identifiers: P51587

Overlapping RefSeq Gene ID 675 matches and has similar biotype of protein_coding

LRG_293 provides a stable genomic reference framework for describing sequence variants for this gene

ENSG00000139618.14

This gene maps to 32,889,611-32,974,403 in GRCh37 coordinates.
View this locus in the GRCh37 archive: ENSG00000139618

Known protein coding

Annotation for this gene includes both automatic annotation from Ensembl and Havana manual curation, see article.

This gene corresponds to the following database identifiers:
Havana gene: OTTHUMG00000017411

overlapping locus [Definitions]

Go to Region in Detail for more tracks and navigation options (e.g. zooming)

此处详细解释见下页

Tab 3: Gene card –Part 1

Gene: BRCA2 ENSG00000139618

Description breast cancer 2 [Source:HGNC Symbol;Acc:[HGNC:1101](#)]

Synonyms XRCC11, FACD, FAD1, PNCA2, FAD, BRCC2, FANCD1, GLM3, BROVCA2, FANCD

Location [Chromosome 13: 32,315,474-32,400,266](#) forward strand.
GRCh38:CM000675.2

About this gene This gene has 7 transcripts ([splice variants](#)), [63 orthologues](#), is a member of [1 Ensembl protein family](#) and is associated with [51 phenotypes](#).

Show transcript table

Summary ?

Name [BRCA2](#) (HGNC Symbol)

CCDS This gene is a member of the Human CCDS set: [CCDS9344.1](#)

UniProtKB This gene has proteins that correspond to the following Uniprot identifiers: [P51587](#)

RefSeq Overlapping RefSeq Gene ID [675](#) matches and has similar biotype of protein_coding

LRG [LRG_293](#) provides a stable genomic reference framework for describing sequence variants for this gene

Ensembl version ENSG00000139618.14

Other assemblies This gene maps to [32,889,611-32,974,403](#) in GRCh37 coordinates.
View this locus in the GRCh37 archive: [ENSG00000139618](#)

Gene type Known protein coding

Annotation method Annotation for this gene includes both automatic annotation from Ensembl and [Havana](#) manual curation, see [article](#).

Alternative genes This gene corresponds to the following database identifiers:
Havana gene: [OTTHUMG00000017411](#)

Annotation Attributes overlapping locus [\[Definitions\]](#)

基因全称

别名

位置，正反链，版本信息

转录本信息

其余信息链接

CCDS 链接

蛋白信息

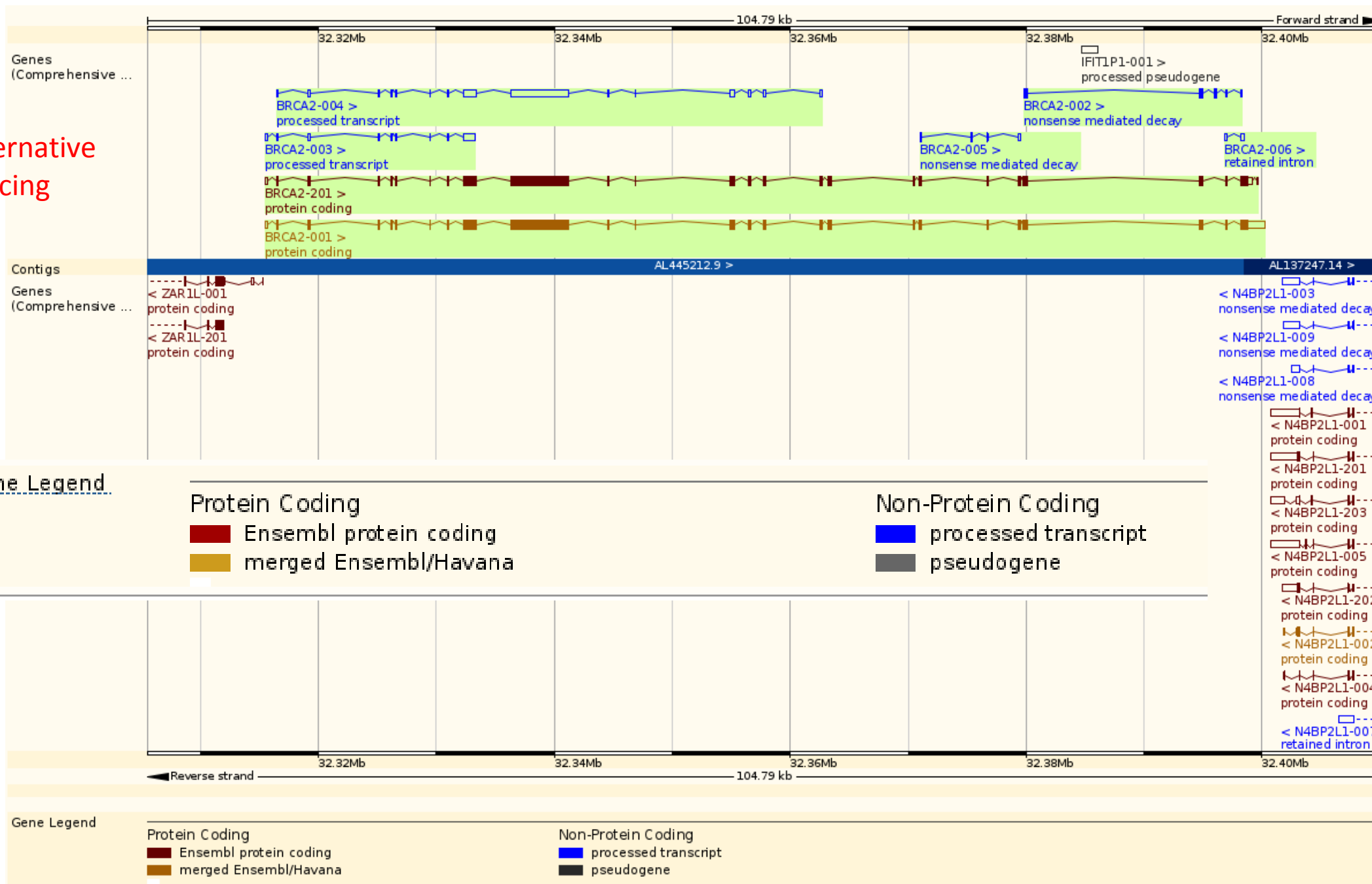
RefSeq信息

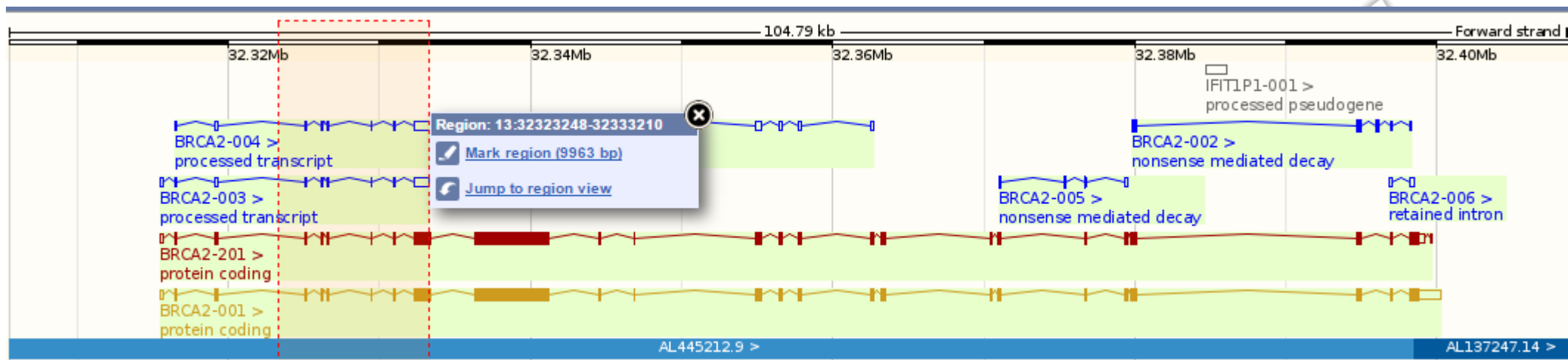
老版本数据库的跳转链接

Havana信息

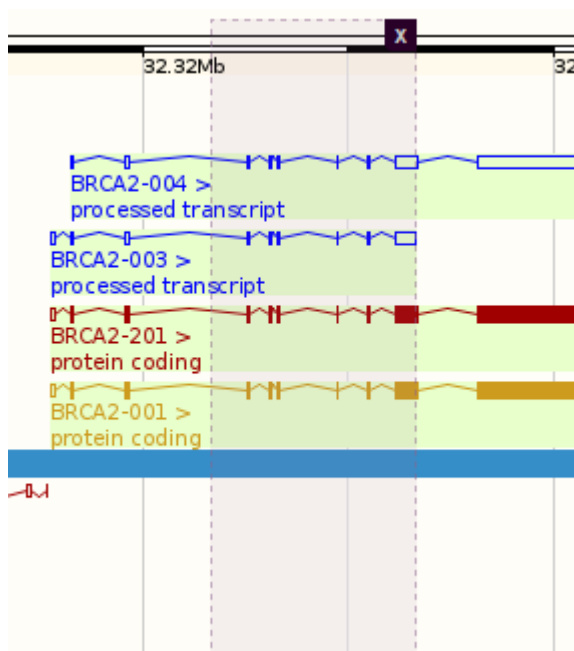
Alternative
splicing

螺旋





Mark this region



整个窗口都是选中区域

图片导出

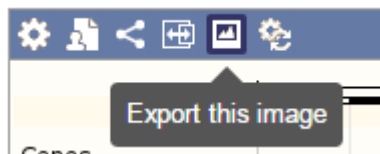


Image download

File name:

Select Format

- ☒ **Journal/report** - High resolution, suitable for printing at A4/letter size
- ☐ **Poster** - Very high resolution, suitable for posters and other large print uses
- ☐ **Presentation** - Saturated image, better suited to projectors
- ☐ **Web** - Standard image, suitable for web pages, blog posts, etc.
- ☐ **PDF file** - Standard image as PDF file
- ☐ **Custom image** - Select from a range of formats and sizes

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Helix 解螺旋
Life

此处详细解释见下页

Gene-based displays

- Summary
- Splice variants
- Transcript comparison
- Supporting evidence
- Gene alleles
- Sequence
 - Secondary Structure
- External references
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- Ontologies
 - GO: Biological process
 - GO: Molecular function
 - GO: Cellular component
- Comparative Genomics
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 - Gene gain/loss tree
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 - Paralogues
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- Phenotype
- Genetic Variation
 - Variant table
 - Variant image
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 - Gene expression
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Tab 3: Gene card Gene based display



Transcript comparison

Key

Exons/Introns	Intron or reference sequence	Translated sequence	UTR
Variations	3 prime UTR	5 prime UTR	Coding sequence
	Downstream	Frameshift	Incomplete terminal codon
	Inframe deletion	Inframe insertion	Initiator codon
	Intronic	Missense	Splice acceptor
	Splice donor	Splice region	Stop gained
	Synonymous		

BRCA2	1	GGGCTTGTGGCGCGAGCTTCTGAAACTAGGCGGCAGAGGCGGAGCCGCTGTGGCACTGCT	60
BRCA2-001	1	GGGCTTGTGGCGCGAGCTTCTGAAACTAGGCGGCAGAGGCGGAGCCGCTGTGGCACTGCT	60
BRCA2-003	1	-----GGCAGAGGCGGAGCCGCTGTGGCACTGCT	60
BRCA2-201	1	-----GTGGCGCGAGCTTCTGAAACTAGGCGGCAGAGGCGGAGCCGCTGTGGCACTGCT	60
BRCA2	61	GCGCCTCTGCTGCGCCTCGGGTGTCTTTTTCGGCGGTTGGGTCGCCGCCGGGAGAAGCGTG	120
BRCA2-001	61	GCGCCTCTGCTGCGCCTCGGGTGTCTTTTTCGGCGGTTGGGTCGCCGCCGGGAGAAGCGTG	120
BRCA2-003	61	GCGCCTCTGCTGCGCCTCGGGTGTCTTTTTCGGCGGTTGGGTCGCCGCCGGGAGAAGCGTG	120
BRCA2-201	61	GCGCCTCTGCTGCGCCTCGGGTGTCTTTTTCGGCGGTTGGGTCGCCGCCGGGAGAAGCGTG	120
BRCA2	121	AGGGGACAGATTTGTGACCGGCGCGGTTTTTGTGAGCTTACTCCGGCCAAAAAAGAACTG	180
BRCA2-001	121	AGGGGACAGATTTGTGACCGGCGCGGTTTTTGTGAGCTTACTCCGGCCAAAAAAGAACTG	180
BRCA2-003	121	AGGGGACAGATTTGTGACCGGCGCGGTTTTTGTGAGCTTACTCCGGCCAAAAAAGAACTG	180
BRCA2-201	121	AGGGGACAGATTTGTGACCGGCGCGGTTTTTGTGAGCTTACTCCGGCCAAAAAAGAACTG	180

Gene-based displays

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Tab 3: Gene card



Marked-up sequence ?

[Download sequence](#) [BLAST this sequence](#)

Exons Cntnap1 exons All exons in this region

Markup loaded

>chromosome:GRCm38:11:101169923:101191324:1

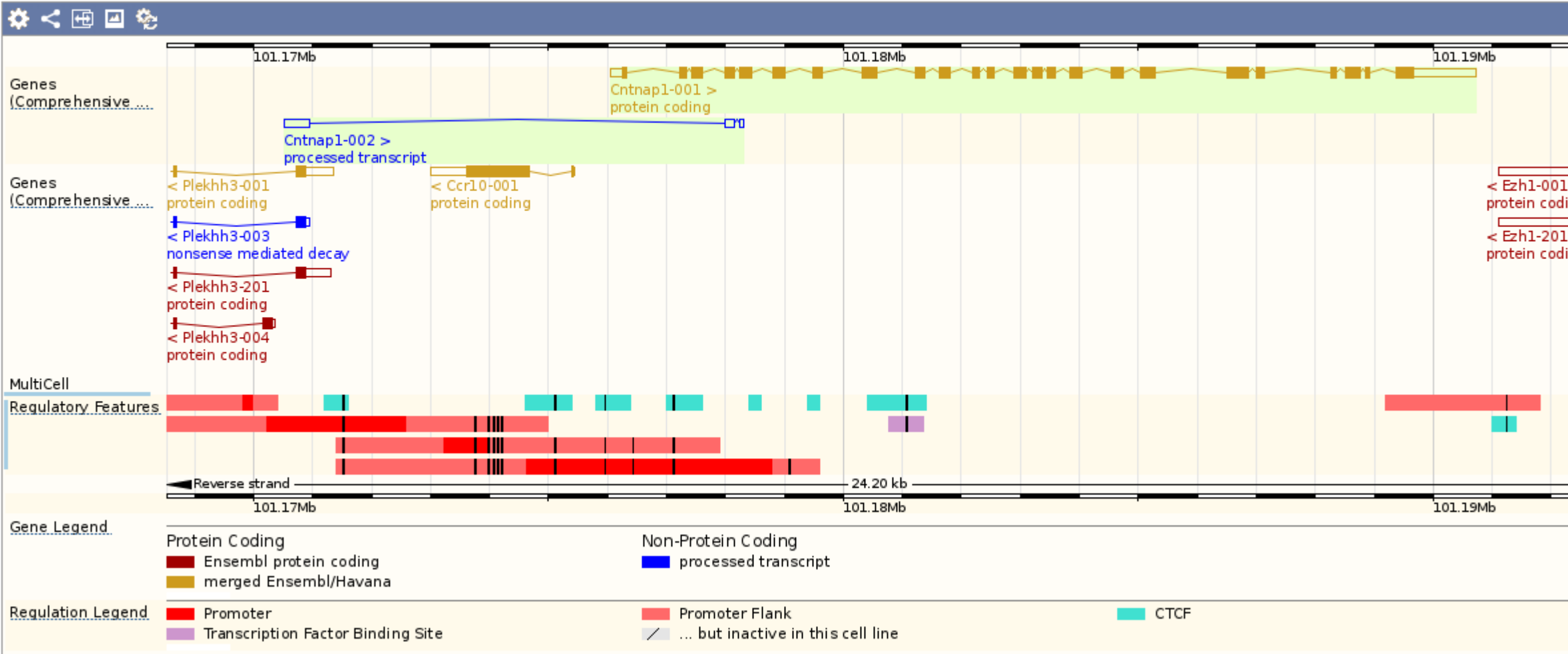
```
GGGACCCCTGAGCGGGGTTGGGGGAAATGGGAGCTGGACCCAGGCCACAGCACACAGAA
GCCTGGTTGCCTCTGGTGGGTAAGGACGTGAGAGAGGTCTCCTCGAAAGATGGAGGTCTT
ACACACCAACTGTGCCTGAGTGTCTCTCTCCCAAAGGAGATTTGATTAAAAAAATTTTC
AGCCACCAAGTCACCCTCACCCATTCCCTCTGGTTTCCCAGGAACCTCACAGAGAAGGC
AGAGAGTTCTGTCTTAAGTGGCGGAATAATGAGAAGTCCTCAAAGGGCCATCGTCCCGAG
CCTTTTCTTTGTGGGGTCCCTGATGCAGACCCCAATTCTAGGATGTCTTTTGGCCCTCGC
CCAGGATACTGCGGTGGGAGCTCCCGCATCTGGCCCCAGTGCTCTTGAAGTGAAGCAGA
GCGAACCCAGGC GTTCCAGCCGCCAGCCGGCCTCGCCACATTCTTGGCTCTGGCGGAG
GCGGAAGGAGACGGGATGGGACGCGGGCCCGGCAGGGAAGGGAGAGGCAGGGCCAGGTTG
GGCCACGCGTGACGCCTCCTCTGCCTCAGGGCCCCAGCAGGTGGACAGCGCCTTGTGTG
CAGCGCGAAGCCCACAGGCGCATCCCTAGTAGGGCTACTTGCCCCCTGGAGCTCCCGGGGC
TCTGGCCCTCAGACAAGAATCTCCCCACATTTGCAGTTGGCCAAGAGGACTGCGTTTGG
CCCAAGTATGGAGCAGGCTCAGGCGTGACGGCCGGTTGTAGTGAGAAAGATTGAAGTCGG
TTCTAAATCCCTGTAGACTTACCCTCCCGCCGCCGCTGGACTCGGGGTCTTAGCTCAA
AGGTCTCGTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCT
GGTGCAGAAGAGTGAAGCCTCGACGGCAGCAGAGAAGCCACCACAATCCCCCGGGGAGAG
GCATCCGGGCGAGCAGCCTGGGAATGGGGGCGCAGGCAGTGTTGCCTGAGCAGCAGGGAA
TCTGAGAAACTGGAGGTACTCCGACCCGAACTGGGTAGAGAAGAGGTGTGACGAGAGGAA
AGAAGGGGGCCGGAAGGAGGCAGTGTCCAGCGCTGAGGTGCAAAAGGACTATCACCAGAG
CCCGCTGGGGGCTCGCTGTCTGGGCCCGGGCAGAGGAGGGGGGAGATAGCGTCCAGGGCC
GGGGGAAGGGAGCCGGGAGCAATGTCCGGAGCTGGGAAGTAGTGTTCGTTGTAGTGTCCA
GCTCTGTAGGAGCAGTGTCTGGTGCAGCAATCGTGGTGGGCGGTTGTCCAAGCTCCAAGG
AGGGTGGTGTCCGGTAGGGCGTCCGGTGGCCGGGGCCCGGGCGGCGCGCTGCGCTCCC
TGCAGCCCCGGGACTGGCGCGTTGGGGGCGCGGGCAGAGTTCACAGACCCCGCCTCCCTC
TCTGGGGGGCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCTCT
```

Tab 3: Gene card



Regulation ?

- Gene-based displays
- Summary
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 - Ontologies



Show All entries		Show/hide columns		Filter	
Reg. region	Analysis	Type	Location	Length (bp)	Sequence (positive strand)
ENSMUSR00000521672	Ensembl Regulatory Build	Promoter	11:101169800-101170001	220	AGTCCCTTGCTTCCCTCCTCCTCCCTCTATATTTATCTGCCCTCTGCCCTCGCCAGGTGC TGGCGCTGCACTGGCTGCGGTGCCGCCGCCGCCACCCAGGAATGGAGCCACCGGACTTG CTGGGGACCCCTGAGCGGGGTGGGGGGAAATGGGAGCTGGACCCAGGCCACAGCACACA GAAGCCTGGTTGCCTCTGGTGG
ENSMUSR00000252025	Ensembl Regulatory Build	Promoter	11:101170200-101172601	2642	CCTCAAAGGGCCATCGTCCCGAGCCTTTTCTTTGTGGGGTCCCTGATGCAGACCCCAATT CTAGGATGCTTTTTTGCCCTCGCCCAGGATACTGCGGTGGGAGCTCCCGCATCTGGCCCC AGTGCTCTTGAACTGAGAGCAGAGCGAACCCAGGCGTTCCAGCCGCCAGCCGGCCTCGC CACATTCTTGGCTCTGGCGGAGGCGGAAGGAGACGGGATGGGACGCGGGCCCCGGCAGGG AAGGGAGAGGCAGGGCCAGGTGGGGCCACGCGTGACGCCCTCTGCCCTCAGGGCCCCCA

CTCF binding site, Promoter, TF binding site.....

GO: Biological process

- Ontologies
 - GO: Biological process
 - GO: Molecular function
 - GO: Cellular component

下载

Show/hide columns					Filter	
Accession	Term	Evidence	Annotation Source	Transcript IDs		
GO:0002175	protein localization to paranode region of axon	IMP	UniProtKB/Swiss-Prot:O54991	ENSMUST00000103109	Search BioMart View on karyotype	
GO:0007010	cytoskeleton organization	IMP	UniProtKB/Swiss-Prot:O54991	ENSMUST00000103109	Search BioMart View on karyotype	
GO:0007155	cell adhesion	IEA	UniProtKB/Swiss-Prot:O54991	ENSMUST00000103109	Search BioMart View on karyotype	
GO:0019227	neuronal action potential propagation	IMP	UniProtKB/Swiss-Prot:O54991	ENSMUST00000103109	Search BioMart View on karyotype	
GO:0030913	paranodal junction assembly	IMP	UniProtKB/Swiss-Prot:O54991	ENSMUST00000103109	Search BioMart View on karyotype	
GO:0031175	neuron projection development	IGI	UniProtKB/Swiss-Prot:O54991	ENSMUST00000103109	Search BioMart View on karyotype	
GO:0048812	neuron projection morphogenesis	IMP	UniProtKB/Swiss-Prot:O54991	ENSMUST00000103109	Search BioMart View on karyotype	
GO:0050884	neuromuscular process controlling posture	IMP	UniProtKB/Swiss-Prot:O54991	ENSMUST00000103109	Search BioMart View on karyotype	
GO:0050885	neuromuscular process controlling balance	IMP	UniProtKB/Swiss-Prot:O54991	ENSMUST00000103109	Search BioMart View on karyotype	

后面的课程中，我会详细解释什么是GO term。

CC. Cellular component

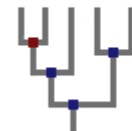
Comparative Genomics

- Genomic alignments
- Gene tree
- Gene gain/loss tree
- Orthologues
- Paralogues
- Ensembl protein families

Comparative Genomics

TCTC---TAGT
TGTCATTGACT
CCGT---GAGC

Alignments



Gene Tree



Orthologues



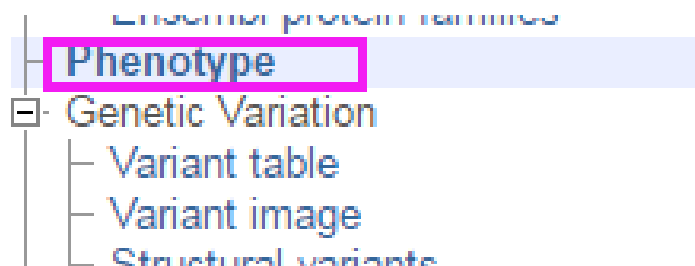
Paralogues



Families

比较生物学信息

表型相关



Phenotype ?

Phenotype(s), disease(s) and trait(s) associated with this gene ENSMUSG00000017167

Filter 					
Phenotype, disease and trait	Source	Study	Genomic Locations	Strain	Allele
cortical actin cytoskeleton organization	GOA	PMID:16551741	 View on Karyotype	C57BL/6J (Unknown)	
neuromuscular process controlling balance	GOA	PMID:16551741	 View on Karyotype	C57BL/6J (Unknown)	
neuromuscular process controlling posture	GOA	PMID:19816196	 View on Karyotype	C57BL/6J (Unknown)	
neuron projection morphogenesis	GOA	PMID:16551741 , PMID:19816196	 View on Karyotype	C57BL/6J (Unknown)	
neuronal action potential propagation	GOA	PMID:19816196	 View on Karyotype	C57BL/6J (Unknown)	
paranodal junction assembly	GOA	PMID:16551741 , PMID:19816196	 View on Karyotype	C57BL/6J (Unknown)	
protein localization to paranode region of axon	GOA	PMID:16551741 , PMID:19816196	 View on Karyotype	C57BL/6J (Unknown)	

No phenotype, disease or trait is associated with variants in this gene.

Phenotype, disease and trait annotations associated orthologues of this gene in other species

Filter 			
Phenotype, disease and trait	Source	Species	Gene
LETHAL CONGENITAL CONTRACTURE SYNDROME 7	MIM morbid	Human (<i>Homo sapiens</i>)	ENSG00000108797 CNTNAP1
Hypomyelination neuropathy - arthrogryposis	Orphanet	Human (<i>Homo sapiens</i>)	ENSG00000108797 CNTNAP1

转录本信息

- Gene card 页面上 Transcripts 项

Filter

Download what you see
Download whole table

Flags



Transcripts

Hide transcript table

Show/hide columns (1 hidden)								Filter	
Name	Transcript ID	bp	Protein	Biotype	CCDS	UniProt	RefSeq	Flags	
BRCA2-001	ENST00000380152	11986	3418aa	Protein coding	CCDS9344	P51587	-	TSL:5	GENCODE basic APPRIS P1
BRCA2-201	ENST00000544455	10984	3418aa	Protein coding	CCDS9344	P51587	NM_000059 NP_000050	TSL:1	GENCODE basic APPRIS P1
BRCA2-002	ENST00000470094	842	186aa	Nonsense mediated decay	-	H0YE37	-	CDS 5' incomplete	TSL:5
BRCA2-005	ENST00000528762	495	64aa	Nonsense mediated decay	-	H0YD86	-	CDS 5' incomplete	TSL:4
BRCA2-004	ENST00000614259	7950	No protein	Processed transcript	-	-	-	TSL:2	
BRCA2-003	ENST00000530893	2011	No protein	Processed transcript	-	-	-	TSL:1	
BRCA2-006	ENST00000533776	523	No protein	Retained intron	-	-	-	TSL:3	

每条转录本记录更加详细的信息记录



Ensembl

BLAST/BLAT | BioMart | Tools | Downloads | Help & Documentation | Blog | Mirrors

Human (GRCh38.p5) | Location: 13:32,330,879-32,339,085 | Gene: BRCA2 | Transcript: BRCA2-001

Search Human...

Login/Register

Transcript-based displays

Summary

Supporting evidence

Sequence

Exons

cDNA

Protein

External References

General identifiers

Oligo probes

Genetic Variation

Variant table

Variant image

Population comparison

Comparison image

Haplotypes

Protein Information

Protein summary

Domains & features

Variants

ID History

Transcript history

Protein history

Transcript: BRCA2-001 ENST00000380152

Description
breast cancer 2 [Source:HGNC Symbol;Acc:HGNC:1101]

Synonyms
XRCC11, FACD, FAD1, PNCA2, FAD, BRCC2, FANCD1, GLM3, BROVCA2, FANCD

Location
Chromosome 13: 32,315,474-32,400,266 forward strand.

About this transcript
This transcript has 27 exons, is annotated with 42 domains and features, is associated with 6770 variations and maps to 60 oligo probes.

Gene
This transcript is a product of gene ENSG00000139618

Hide transcript table

Show/hide columns (1 hidden) | Filter

Name	Transcript ID	bp	Protein	Biotype	CCDS	UniProt	RefSeq	Flags
BRCA2-001	ENST00000380152	11986	3418aa	Protein coding	CCDS9344	P51587	-	TSL:5 GENCODE basic APPRIS P1
BRCA2-201	ENST00000544455	10984	3418aa	Protein coding	CCDS9344	P51587	NM_000059 NP_000050	TSL:1 GENCODE basic APPRIS P1
BRCA2-002	ENST00000470094	842	186aa	Nonsense mediated decay	-	H0YE37	-	CDS 5' incomplete TSL:5
BRCA2-005	ENST00000528762	495	64aa	Nonsense mediated decay	-	H0YD86	-	CDS 5' incomplete TSL:4
BRCA2-004	ENST00000614259	7950	No protein	Processed transcript	-	-	-	TSL:2
BRCA2-003	ENST00000530893	2011	No protein	Processed transcript	-	-	-	TSL:1
BRCA2-006	ENST00000533776	523	No protein	Retained intron	-	-	-	TSL:3

Summary

84.79 kb

Forward strand

Configure this page

Add your data

Export data

Share this page

Bookmark this page

检索-疾病名称



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Human (GRCh38.p2) ▼

Current selection:

< all Species
Only searching Human

Restrict category to:

Gene	258
Transcript	267
Variation	820
Phenotype	100
Somatic Mutation	83536
Protein Domain	4
Protein Family	20

Per page:

10 25 50 100

Layout:

Standard Table

Tip:

Only searching Human liver failure

85005 results match liver failure when restricted to species: Human ✕

Did you mean... ▼

[liver failure acute infantile \(Human Phenotype\)](#)
Human Phenotype
Liver failure acute infantile .

[liver failure, infantile, transient \(Human Phenotype\)](#)
Human Phenotype
Liver failure, infantile, transient .

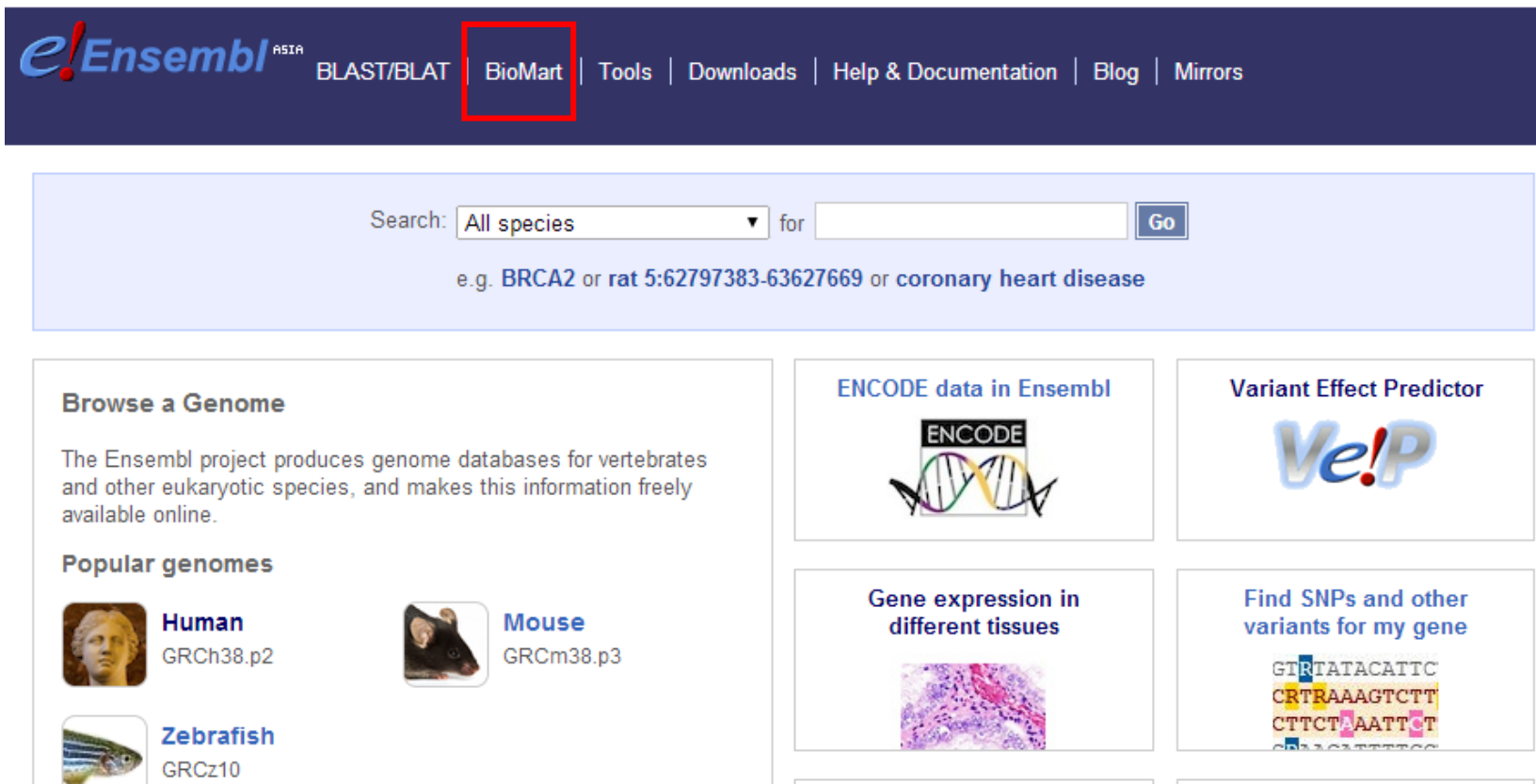
[renal tubulopathy - encephalopathy - liver failure \(Human Phenotype\)](#)
Human Phenotype
Renal tubulopathy - encephalopathy - liver failure .

[infantile liver failure syndrome 2 \(Human Phenotype\)](#)
Human Phenotype
Infantile liver failure syndrome 2 .

[infantile liver failure syndrome 1 \(Human Phenotype\)](#)
Human Phenotype
Infantile liver failure syndrome 1 .

其他工具-BioMart

- BioMart是一个集成的数据接口，可以通过其实现批量的数据检索



The screenshot shows the Ensembl BioMart website. At the top, the Ensembl logo is followed by navigation links: BLAST/BLAT, **BioMart** (highlighted with a red box), Tools, Downloads, Help & Documentation, Blog, and Mirrors. Below the navigation bar is a search section with a dropdown menu set to 'All species', a text input field, and a 'Go' button. Below the search bar, an example query is provided: 'e.g. BRCA2 or rat 5:62797383-63627669 or coronary heart disease'. The main content area is divided into several sections:

- Browse a Genome**: A section describing the Ensembl project's goal to produce genome databases for vertebrates and other eukaryotic species, making this information freely available online.
- Popular genomes**: A section featuring three popular genomes:
 - Human**: GRCh38.p2, accompanied by a classical statue icon.
 - Mouse**: GRCm38.p3, accompanied by a mouse icon.
 - Zebrafish**: GRCz10, accompanied by a zebrafish icon.
- ENCODE data in Ensembl**: A section featuring the ENCODE logo and a DNA double helix icon.
- Variant Effect Predictor**: A section featuring the 'Ve!P' logo.
- Gene expression in different tissues**: A section featuring a histology image of tissue.
- Find SNPs and other variants for my gene**: A section featuring a DNA sequence snippet:


```

      GTTATACATTG
      CRTAAAGTCTT
      CTTCTAATTCT
      CAGCTTTCG
      
```



Dataset [None selected]	- CHOOSE DATABASE - ▾ - CHOOSE DATABASE - Ensembl Genes 84 Ensembl Variation 84 Ensembl Regulation 84 Vega 64
-----------------------------------	--

其他工具-BioMart



Database-filter-output

e!Ensembl ASIA [BLAST/BLAT](#) | [BioMart](#) | [Tools](#) | [Downloads](#) | [Help & Documentation](#) | [Blog](#)

[New](#) [Count](#) [Results](#) [★ URL](#) [XML](#) [Perl](#) [Help](#)

Dataset Homo sapiens genes (GRCh38.p2)	Ensembl Genes 80 ▼
Filters [None selected]	
Attributes Ensembl Gene ID Ensembl Transcript ID	
Dataset [None Selected]	Homo sapiens genes (GRCh38.p2) ▼

Dataset

Filters

[None selected]

Attributes

Ensembl Gene ID
Ensembl Transcript ID

Dataset

[None Selected]

Please select columns to be included in the output and hit 'Results' when ready

Missing non coding genes in your mart query output, please check the following [FAQ](#)

- ☒ Features
- ☐ Structures
- ☐ Homologues
- ☐ Variant (Germline)
- ☐ Variant (Somatic)
- ☐ Sequences

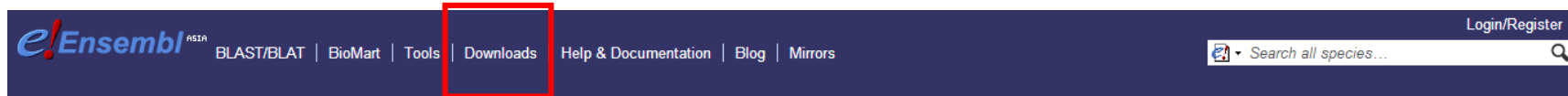
☐ GENE:

☐ EXTERNAL:

☐ PROTEIN DOMAINS AND FAMILIES:



数据库下载



Download a sequence or region

Export data

CAGAATGAT
AAATGTTCT
AAAGAAGCA
CTGTCTATGC
ATAAAGAA
AGTGATACT



Click on the 'Export data' button in the lefthand menu of most pages to export:

- FASTA sequence
- GTF or GFF features

...and more!

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You may find exploring this web-based query tool easier than extracting information direct from our databases.

Fetch data programmatically



Write your own Perl scripts to retrieve small-to-medium datasets. All our data, as well as added functionality, is available through the Ensembl Perl API.

Use the API to retrieve gene and transcript sets, fetch alignments between sequences, compare allele frequencies and much more!

You can also use our [REST API](#) to retrieve data to process in the programming language of your choice.

Download databases & software



All of our data and software, including pipelines and web code, is available free.

- [Download data via FTP](#)
- [Ensembl pipeline on GitHub](#)
- [Set up your own Ensembl website](#)

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<http://asia.ensembl.org/Multi/Tools/AssemblyConverter>

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Web Tools

- Web Tools
 - BLAST/BLAT
 - Variant Effect Predictor
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Assembly Converter

New Assembly Converter job:

This online tool currently uses [CrossMap](#), which supports a limited number of formats (see our online documentation for [details of the individual data formats](#) listed below). CrossMap also discards metadata in files, so track definitions, etc, will be lost on conversion.

Important note: CrossMap converts WIG files to BedGraph internally for efficiency, and also outputs them in BedGraph format.

Input

Species:	Human (Homo sapiens) ▾
Assembly mapping:	GRCh37 -> GRCh38 ▾
Name for this data (optional):	
Input file format:	BED ▾
Either paste data:	

总结

- Ensembl提供了非常多物种的基因注释信息
- 同时，对genomic location→ gene→transcript 进行了多层面的链接
- 提供了Biomart批量化下载工具

Thanks～